Development and robustness: slower trading helped, but did not establish a durable edge

Final report 1 of 3 | Restarted crypto research | 1 January 2022-31 December 2023

Executive Summary

- Only **106,302 of 1,117,935 paths (9.51%)** had positive compounded net return at 20 bps. The full-grid median was **-71.03%**, versus **-7.89% before modeled fees**. Fast turnover was a major obstacle, not the only one.
- Slower trend definitions had better development distributions: median net returns were **+4.05% long/cash** and **+22.04% theoretical long/short**. These are unequal family grids, not evidence that trend universally dominates cross-sectional strategies.
- The apparent edge was uneven. **Every long/cash path lost in 2022 and gained in 2023**; only **39.15% of long/short paths** were positive in both years. Basket and event concentration mattered.
- This stage supported **development robustness checks before selection**, not a discovery claim. All unsuccessful configurations remain in the archive; the 169-definition space evolved in earlier retrospective research.

What was restarted

The restart retained the cryptocurrency history and full configuration space, but did not import previous winning baskets, survivor filters or selection conclusions. It could not erase earlier inspection of those historical years.

The complete study covered **6,615 eight-coin baskets × 169 definitions**, or **1,117,935 primary basket-configuration pairs**. Seven fee/delay scenarios produced **7,825,545 result rows**. All were evaluated; **1,000 genuine all-cash paths had undefined Sharpe**, not zero Sharpe. Every basket had at least one positive development result, which is unsurprising after a large correlated search.

Development was 2022-2023. Validation was reserved as 2024-2025 and 2026 as a later evaluation; neither was used in the development evidence presented here. Reports 2 and 3 cover those subsequent stages.

The basket rule used historical liquidity within an inherited pool

At the 1 January 2022 decision, coins were ranked by **median daily USDT quote volume across 90 observations**, with endpoints from **3 October through 31 December 2021** and a **24-hour information lag**. This used the inherited historical top-30 research universe, not an unrestricted census of every exchange listing. The restarted basket campaign used ranks **1-22** only, with no reconstitution.

BTC and ETH were fixed. Each basket took two coins from ranks 3-9, two from 10-16 and two from 17-22: **C(7,2) × C(7,2) × C(6,2) = 6,615** baskets. Every signal was recomputed inside its own eight-coin basket; thirty-coin portfolio weights were not sliced.

Rank is historical liquidity rank; volume is millions of USDT per day. Quote volume measures activity, not a demonstrated executable spread or trading capacity.

Fixed slots

Rank	Pair	Median volume (USDT m)
1	BTCUSDT	2384.13
2	ETHUSDT	1724.40

Tier 1: choose two from ranks 3-9

Rank	Pair	Median volume (USDT m)
3	SHIBUSDT	718.95
4	BNBUSDT	605.06
5	SOLUSDT	463.10
6	XRPUSDT	385.78
7	DOTUSDT	353.16
8	SANDUSDT	303.11
9	LUNAUSDT	285.00

Tier 2: choose two from ranks 10-16

Rank	Pair	Median volume (USDT m)
10	AVAXUSDT	284.44
11	DOGEUSDT	272.77
12	ADAUSDT	261.92
13	FTMUSDT	244.35
14	TRXUSDT	228.72
15	MATICUSDT	223.37
16	MANAUSDT	208.63

Tier 3: choose two from ranks 17-22

Rank	Pair	Median volume (USDT m)
17	FILUSDT	203.47
18	GALAUSDT	183.98
19	LINKUSDT	140.30
20	AXSUSDT	121.01
21	LTCUSDT	116.42
22	VETUSDT	103.97

LUNA denotes the **legacy Terra/LUNC economic slot, not Terra 2.0**. A suspension or unavailable slot remained cash under the preserved lifecycle policy; it did not trigger replacement by a current survivor.

Six families, with different economic exposures

Momentum and reversal: 78 definitions each. The lookbacks were 1, 2, 4, 8, 12, 24, 48, 72, 168, 336, 672, 1,440 and 2,160 hours; rebalancing was every 1, 4, 12, 24, 72 or 168 hours. Cross-sectional trailing-return ranks were demeaned and scaled to full-L1 gross exposure one, with dollar neutrality. Reversal used the opposite sign. Dollar neutrality does not ensure BTC-beta neutrality.

Buffered momentum: four definitions. The 14/28-day lookbacks crossed daily/weekly decisions. The entire rebalance was skipped if proposed full-L1 target change was below 0.25; otherwise the full target was executed. Four matching unbuffered momentum definitions were already in the 78-case grid.

Time-series trend: four long/cash and four theoretical long/short definitions. The same 14/28-day lookbacks crossed daily/weekly decisions. Each coin had an unscaled 1/8 slot: positive signal meant +1/8 for long/cash, otherwise cash; long/short used sign(trailing return)/8. Remaining active positions were not rescaled.

One fixed BTC-factor residual reversal. For seven non-BTC coins, monthly formation used 90 daily observations ending 24 hours before refit: earliest 30 to estimate BTC return beta, subsequent 60 endpoints to calibrate log-price spread mean/standard deviation. Eligibility required beta in [0.1,5], positive variance, Holm-adjusted ADF significance at 5%, and AR(1) half-life 1-20 days. Coefficients stayed frozen during the month.

Residual entries required $2 \le |z| < 4$, toward the mean. Exits used the ±0.5 mean boundary, adverse ±4 stop, or 20 calendar days held; no same-day reentry. Month-start forced cash/refit, entries from the next day. Each spread had 1/7 of capital including its BTC hedge; hedges were netted before costs. Ineligible spreads stayed cash. This model was not necessarily dollar- or beta-neutral.

Common execution and accounting

Hourly Binance spot-USDT history began with warm-up on 1 October 2021. Each evaluated period started in cash and paid entry and terminal liquidation charges. Elapsed-time schedules and the inherited Monday epoch were preserved; weekly is not every seventh available row.

Primary costs were **20 bps one way × full-L1 target changes**, with 0/5/10/15/40 bps sensitivities. **Lag1 means an idealized observed-close fill, not one hour of elapsed execution latency**; lag2 adds one hour. Halt bridges, missingness, forced exits and partial event accounting were preserved. There was no silent filling or sample shortening.

Trend had better development medians, but family grids were not comparable experiments

At 20 bps, unbuffered momentum and reversal had strongly negative median paths. Trend and buffered momentum used only four slow definitions each, compared with 78 definitions in each unbuffered family. Thus the broad family comparison mixes signal form, cadence, lookback and exposure. It cannot establish a universal ranking of trading styles.

2022-2023 development: distribution medians across each family's complete grid

Family	Positive	Gross return	Net return	Sharpe	Max DD
Momentum	9.51%	-1.44%	-60.86%	-1.53	72.36%
Reversal	2.62%	-21.03%	-83.81%	-2.31	85.73%
Buffered momentum	40.51%	24.51%	-6.99%	0.02	39.80%
Trend long/cash	56.30%	17.53%	4.05%	0.25	45.84%
Trend long/short	65.59%	58.16%	22.04%	0.46	58.26%
BTC residual reversal	11.34%	-0.46%	-0.58%	-0.64	0.81%

All return numbers are compounded **two-year returns**, not annual returns. Max DD is nonnegative loss depth. Turnover is annualized **full-L1 target-change multiples**, including entry/liquidation, not conventional half-L1 turnover or actual drift-adjusted traded notional. Column medians may correspond to different individual paths.

Family median annual full-L1 turnover was **268.69× momentum/reversal**, **62.26× buffered momentum**, **20.75× long/cash**, **41.56× long/short**, and **0.29× residual reversal**.

The residual model's small median loss and low turnover reflect sparse eligibility/trading, not a low-risk established edge. Its all-cash cases retain undefined Sharpe. The theoretical short books omit borrow/funding/margin constraints. Long/cash carries variable market participation and cash, so equal gross/net returns do not imply equal risk.

Read risk alongside return

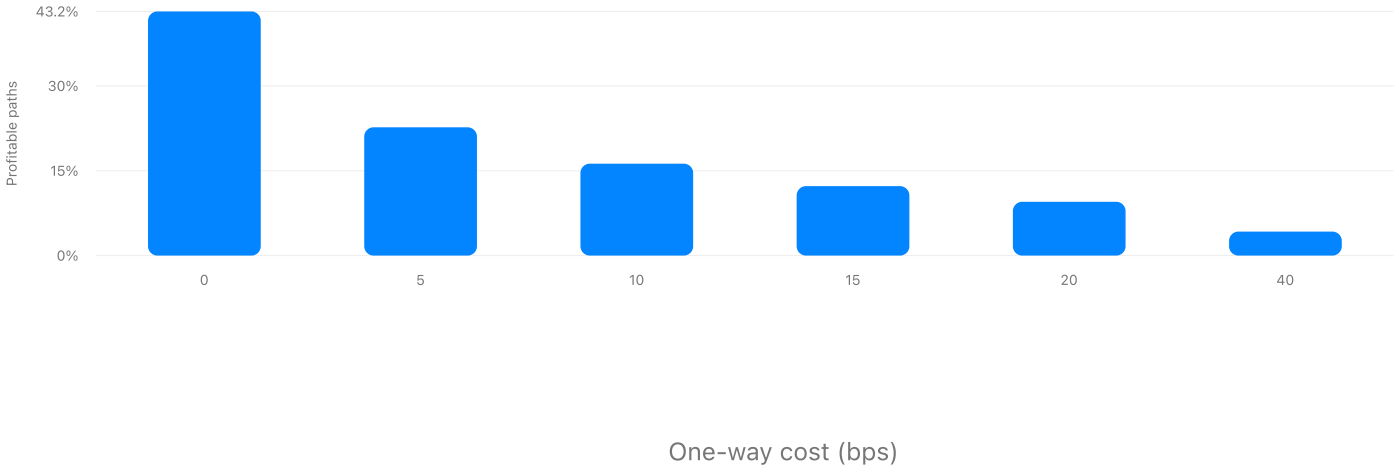
Across the entire unequal grid, the median CAGR was **-46.18%**, annual volatility **27.15%**, Max DD **76.85%**, and longest underwater span **708.6 days**. The median net Sharpe was **-1.77** and annualized full-L1 turnover **189.23×**. These summarize individual research paths, not an implementable aggregate portfolio.

The archive also retains Sortino, Calmar, expected shortfall, worst day/week, ulcer index, win rate, gain/loss ratio, time underwater, exposure, and BTC alpha/beta. No single maximum or aggregate score substitutes for these risks.

Fees destroyed much of the broad grid's already weak economics

At zero modeled fees, **43.17%** of development paths were profitable. At 20 bps that fell to **9.51%**, and at 40 bps to **4.24%**. The zero-cost median was already negative, so the problem was not solely execution costs. Gross return minus a simple percentage fee is not the net compounded result.

Full-grid profitable breadth under sampled fees



All retained fee/delay scenarios

Scenario	Profitable	Median return	Median Sharpe
cost00_lag1	43.17%	-7.89%	-0.01
cost05_lag1	22.68%	-36.87%	-0.70
cost10_lag1	16.25%	-51.34%	-1.09
cost15_lag1	12.28%	-62.42%	-1.44
cost20_lag1	9.51%	-71.03%	-1.77
cost40_lag1	4.24%	-87.46%	-3.03
cost20_lag2	9.60%	-70.80%	-1.75

Slow schedules helped, but weekly was not automatically safer

The pre-existing four-clock intersection compared five nonfactor families at 14/28-day lookbacks and daily/weekly decisions on all 6,615 baskets. This removed unequal lookback/cadence coverage, but not exposure or allocation differences.

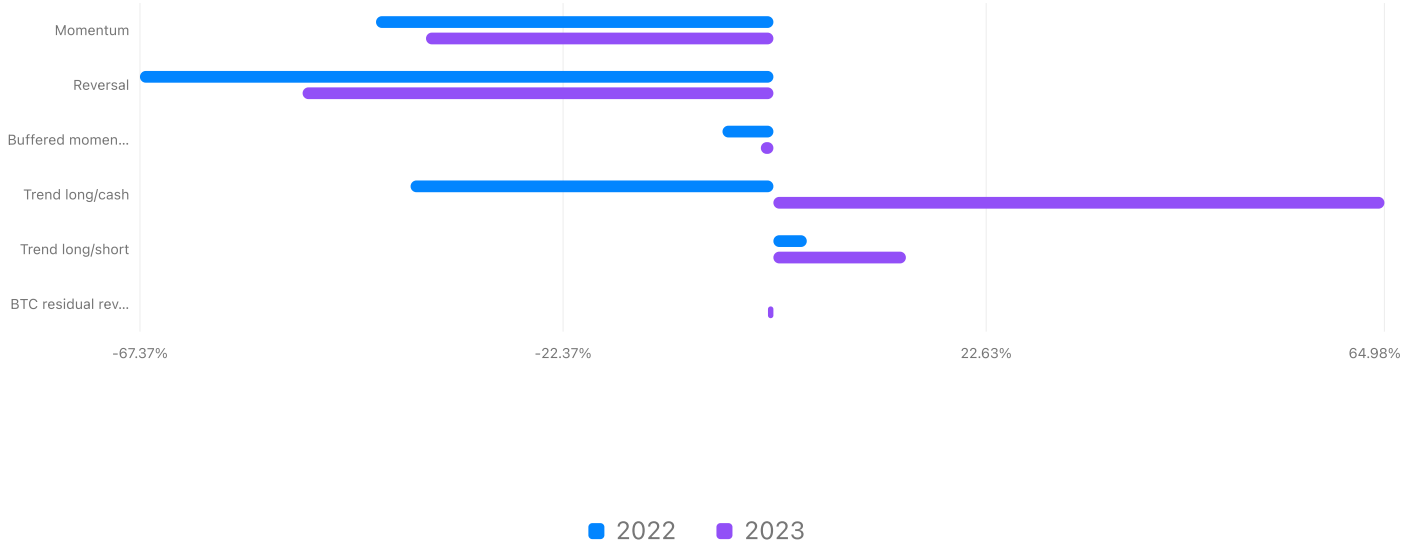
For 14-day long/cash, mean development return was **+1.70% daily versus -25.72% weekly**; for 28-day long/cash it was **+17.33% versus +26.81%**. These are means across matched basket sets, not portfolio returns. Thus “slow reduces costs” was supported, while “weekly always improves outcomes” was not.

The whole-target buffer improved paired median net return by **1.09 pp (14-day daily)** and **3.05 pp (28-day daily)**, but had approximately zero median effect for the two weekly clocks. It was a specific turnover intervention, not a universal signal improvement. The extra-hour scenario retained the same overall weak median economics.

A two-year profit did not imply success in both years

All **26,460 long/cash basket-definition pairs lost in 2022 and gained in 2023**. Only **10,358 of 26,460 (39.15%) long/short pairs** were positive in both. Across every family, just **44,982 of 1,117,935 paths (4.02%)** were positive in both years. These highly correlated counts are descriptive, not independent replications.

Annual median net returns by family, 20 bps



Same-pair annual sign transitions

Family	Both positive	Only 2022	Only 2023	Neither
Momentum	4.78%	7.03%	9.26%	78.94%
Reversal	0.69%	5.35%	3.33%	90.63%
Buffered momentum	24.22%	14.07%	22.82%	38.89%
Trend long/cash	0.00%	0.00%	100.00%	0.00%
Trend long/short	39.15%	15.49%	22.06%	23.31%
BTC residual reversal	0.00%	0.00%	11.34%	88.66%

Annual figures were **slices of the same continuous development path**, not separate annual re-runs. No artificial January cash reset, entry or liquidation fee was inserted; natural scheduled trades/refits and their actual fees remained at their original endpoints.

Within-basket configuration Sharpe rank correlation between the two years had median **0.867** across 6,615 baskets. That high pooled rank correlation can reflect stable separation of very expensive versus slow definitions; it does not demonstrate persistence among profitable leaders or identify a robust parameter winner.

Nearby baskets were often similar; nearby parameters could still change the result

Parameter neighbors changed one adjacent declared lookback or rebalancing level, preserving the other coordinate, family, phase and buffer. Basket neighbors replaced exactly one nonfixed coin within its liquidity tier: **28 neighbors per basket**, sharing seven of eight coins. These were not independent tests.

The table conditions its neighbor-positive share on an already-positive center with finite neighbors; it does not eliminate failures from the full grid. Absolute return changes are medians across all centers and do not cancel opposite signed changes.

Declared local robustness, without selecting a plateau

Family	Param. positive	Basket positive	Param. |ΔR|	Basket |ΔR|
Momentum	50.03%	67.43%	14.67 pp	6.87 pp
Reversal	29.79%	54.56%	8.16 pp	4.63 pp
Buffered momentum	66.08%	73.15%	18.56 pp	14.85 pp
Trend long/cash	55.32%	89.71%	26.48 pp	8.12 pp
Trend long/short	62.37%	92.27%	61.17 pp	19.78 pp
BTC residual reversal	undefined	64.29%	undefined	0.21 pp

For positive momentum centers, the **mean fraction** of profitable parameter neighbors was **50.03%**; for reversal it was **29.79%**. The corresponding basket-neighbor fractions were **67.43%** and **54.56%**. Trend's basket fractions were higher, but the theoretical long/short median absolute parameter change was **61.17 percentage points** of two-year return. A profitable neighborhood was not an invariant signal.

The factor model has no parameter neighbor because it has only one declared definition; that comparison is **unavailable, not zero**. Near-zero signed perturbation summaries were not treated as proof of robustness. No neighbor threshold or candidate was selected in this development report.

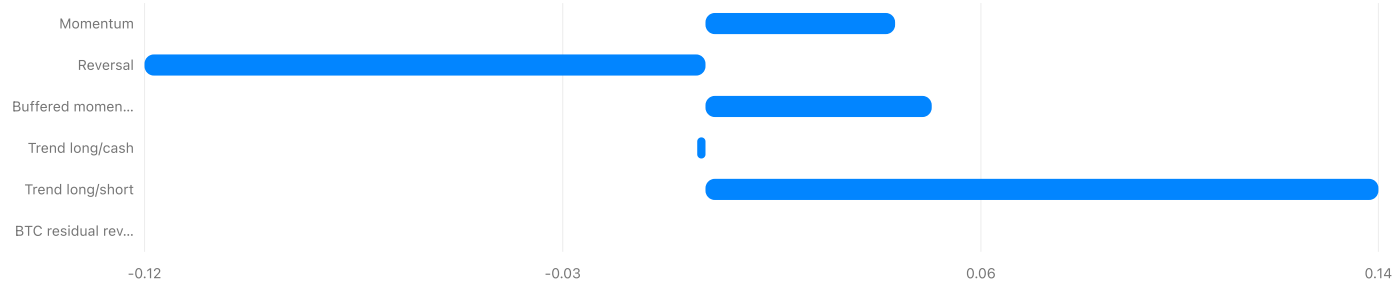
Legacy LUNA exposure was a material event and basket dimension

There were **1,890 baskets with legacy LUNA/LUNC** and **4,725 without**. Long/short trend's median development return was **+52.33% with LUNA versus +14.36% without**. Reversal was worse with LUNA: **-92.52% versus -73.20%**. Other basket memberships also changed, so these are membership-conditioned associations, not causal coin-removal effects.

All-family basket composition comparison, 20 bps

Family	With LUNA return	Without LUNA return	With LUNA Sharpe	Without LUNA Sharpe
Momentum	-59.00%	-61.51%	-0.91	-1.80
Reversal	-92.52%	-73.20%	-2.14	-2.46
Buffered momentum	-3.88%	-8.30%	0.12	-0.04
Trend long/cash	4.73%	3.74%	0.25	0.24
Trend long/short	52.33%	14.36%	0.65	0.41
BTC residual reversal	-1.05%	-0.48%	-0.55	-0.78

Legacy LUNA net dollar contribution in May 6-20, 2022



The event-window mean LUNA contribution was **+$0.1448** for long/short trend and **-$0.1207** for reversal, per original $1 continuous NAV and averaged over **all family paths**, including structural zeros when LUNA was absent. This is reconciled traded-coin P&L accounting, not a counterfactual return after deleting the event. The event window overlaps 2022 and the full path and must not be added twice.

The forced exit used the last **00:40 UTC partial quote on 13 May**, in the 01:00 accounting bucket: 40 minutes of LUNA exposure and possible temporary orphan hedge. This idealized liquidation does not establish executable short profit, borrow availability, or treatment of Terra 2.0/airdrop liabilities.

What these development results did and did not establish

Measurement. Full-period return, elapsed CAGR, drawdown, fees, exposure and turnover used every requested accounting interval. Sharpe was annualized from complete contiguous UTC daily net returns using 365 days; a constant/all-cash series had undefined Sharpe. BTC alpha/beta used the same daily benchmark convention and did not turn a neutral-dollar portfolio into a beta-neutral one. Drawdowns were observed-endpoint measures; intrahour lows and halt interiors were not observed.

Robustness coverage. The retained work included all seven full-period scenarios; both annual slices; exact nonnegative compounded fee roots at both delays; coin/day/month concentration; traded-coin attribution for the full path, each year and the predeclared May event window; matched clocks/buffers; and local parameter/basket perturbations. Finite fee roots were solved from compounded hourly net returns, not gross-return/turnover division. Gross-loss, flat zero-turnover, infinite-tolerance and unavailable states stayed distinct.

Reliability and limits. Complete work-key/scalar identities and saved-file checksums were audited. Independent public-engine reconstruction covered 225 basket-case samples and 1,575 financial scenarios, not every hourly path. Ranking used past liquidity, but inherited universe/source selection remained. The 169-case space itself had evolved retrospectively; correlated basket/parameter searches were not search-adjusted discoveries.

The execution model excluded funding, borrow, margin constraints, order-book depth, slippage, market impact and ordinary drift-maintenance trades. Historical volume did not validate capacity. No prices or strategies were rerun to produce this consolidated report; it reuses accepted saved evidence.

Development-stage conclusion

The defensible development finding was **a cost-and-stability problem**, with some slower configurations worth testing under a rule frozen before validation. It was not “choose the highest development Sharpe” or “trend is proven superior.”

Report 2 follows the development-only selection rule, validation failure diagnosis, reference comparisons and Track B shortlist. Report 3 covers the partial-year 2026 evaluation and subsequent exploratory screening diagnostics. All earlier technical outputs remain archived, but these three reports are the final reading set.

Companion evidence: compact tables alongside this report retain the displayed denominators and units; the source archive retains every unsuccessful configuration, complete metric tables, accounting diagnostics, original notebooks and provenance receipts. No strategy, selection, basket or result has changed in this presentation-only consolidation.